

[2.2] Vanilla Policy Gradient

David Quarel

ILIAD Intensive

October 6, 2026

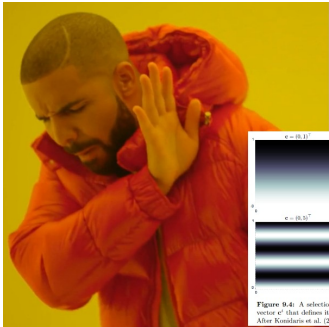
- Tabular RL assumes that sweeping through all state-action pairs is tractable

Beyond Tabular Learning

- Tabular RL assumes that sweeping through all state-action pairs is tractable
- What about large/continuous state spaces?
 - State aggregation?
 - Parameterised policy π_{θ} , learn best θ ?
 - Craft a heuristic by hand?
- In general, would like the agent to learn useful features for us

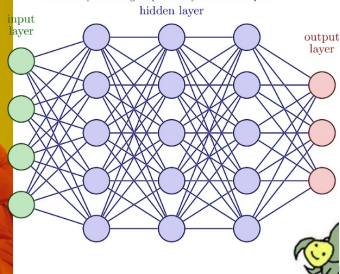
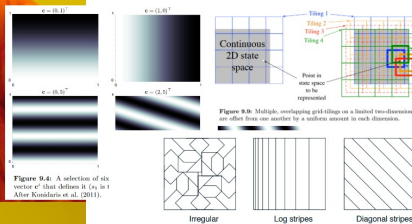
Beyond Tabular Learning

- Tabular RL assumes that sweeping through all state-action pairs is tractable
- What about large/continuous state spaces?
 - State aggregation?
 - Parameterised policy π_{θ} , learn best θ ?
 - Craft a heuristic by hand?
- In general, would like the agent to learn useful features for us
 - Something deep learning excels at!



9.5 Feature Construction for Linear Methods

- 9.5.1 Polynomials
- 9.5.2 Fourier Basis
- 9.5.3 Coarse Coding
- 9.5.4 Tile Coding
- 9.5.5 Radial Basis Functions



Difficulties with using Neural Networks for RL

- Neural networks expect to be trained in a supervised learning fashion, with batches of data fed in, loss computed, and gradients backpropagated.

Difficulties with using Neural Networks for RL

- Neural networks expect to be trained in a supervised learning fashion, with batches of data fed in, loss computed, and gradients backpropagated.
- We need a way to obtain a gradient somehow from trajectories, and then use this gradient to update the policy.

Difficulties with using Neural Networks for RL

- Neural networks expect to be trained in a supervised learning fashion, with batches of data fed in, loss computed, and gradients backpropagated.
- We need a way to obtain a gradient somehow from trajectories, and then use this gradient to update the policy.
- Interaction with environment is *NOT* i.i.d, which supervised learning assumes, so we need to run many trajectories in parallel.

Difficulties with using Neural Networks for RL

- Neural networks expect to be trained in a supervised learning fashion, with batches of data fed in, loss computed, and gradients backpropagated.
- We need a way to obtain a gradient somehow from trajectories, and then use this gradient to update the policy.
- Interaction with environment is *NOT* i.i.d, which supervised learning assumes, so we need to run many trajectories in parallel.
- Our growth truth signal is if the policy obtains large returns during training.

- **Action space** $\mathcal{A}$ is still discrete, and finite
- **State space** $\mathcal{S}$ is continuous (a subspace of $\mathbb{R}^n$), and each state is a vector $\mathbf{s} \in \mathcal{S}$.
- Environment is **episodic**, and agent always starts at/near **starting state** $\mathbf{s}_0$. We generate a full episode, and then afterwards learn. No discount ($\gamma = 1$).
- Define measure of performance (**joy**) as

$$J(\boldsymbol{\theta}) := V_{\pi_{\boldsymbol{\theta}}}(\mathbf{s}_0)$$

- Policy $\pi_{\boldsymbol{\theta}} \equiv \pi(\cdot | \mathbf{s}; \boldsymbol{\theta})$ is a conditional probability distribution $\pi_{\boldsymbol{\theta}} : \mathcal{S} \rightarrow \Delta\mathcal{A}$ over actions, mapping a state to a probability distribution over actions.
- Gradient $\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}$ is well-defined.
- Choose $\boldsymbol{\theta}$ to maximize joy $J(\boldsymbol{\theta})$ via *gradient ascent*

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$$

Softmax policy

- In practice, $\pi_{\theta} : \mathcal{S} \rightarrow \Delta\mathcal{A}$ is a neural network $\mathbf{h}$, mapping a state vector $\mathbf{s}$ to a vector of logits, one for each action $\mathcal{A} = \{a_1, \dots, a_n\}$.

$$\mathbf{h}(\mathbf{s}; \theta) = [h_1(\mathbf{s}; \theta), \dots, h_n(\mathbf{s}; \theta)]$$

from which the action is sampled by softmax'ing the logits

$$\pi(a_i | \mathbf{s}; \theta) := \frac{\exp h_i(\mathbf{s}; \theta)}{\sum_j \exp h_j(\mathbf{s}; \theta)}$$

- More principled than ϵ -greedy, as we get exploration for free, and exploratory actions are not selected at random, but with a bias towards actions believed to be good.

Policy Gradient Framework

- Assume episodic environment, episode length (horizon) T , no discount $\gamma = 1$.
- Assume fixed starting state s_0 . Markov environment $\mu(s'|s, a)$.
- Define $J(\theta) = V_{\pi_\theta}(s_0) = \mathbb{E}_{\tau \sim \pi_\theta}[G(\tau)]$.
- Let $\tau = s_0, a_0, r_1, s_1, \dots, s_{T-1}, a_{T-1}, r_T, s_T$ denote a trajectory
- $G(\tau) = \sum_{t=1}^T r_t$ is the undiscounted return for trajectory τ .

$$\begin{aligned}\Pr(\tau|\theta) &= \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \\ &= \pi_\theta(a_0|s_0) \mu(s_1|s_0, a_0) \dots \pi_\theta(a_{T-1}|s_{T-1}) \mu(s_T|s_{T-1}, a_{T-1})\end{aligned}$$

is the prob. of sampling trajectory τ from enviro. μ given params. θ .

Want to compute $\nabla_\theta J(\theta)$.

Policy Gradient Framework

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)]$$

Policy Gradient Framework

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)] \\ &= \nabla_{\theta} \sum_{\tau} \text{Pr}(\tau|\theta) G(\tau)\end{aligned}$$

Policy Gradient Framework

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)] \\ &= \nabla_{\theta} \sum_{\tau} \text{Pr}(\tau|\theta) G(\tau) \\ &= \sum_{\tau} \nabla_{\theta} \text{Pr}(\tau|\theta) G(\tau)\end{aligned}$$

Policy Gradient Framework

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)] \\ &= \nabla_{\theta} \sum_{\tau} \text{Pr}(\tau|\theta) G(\tau) \\ &= \sum_{\tau} \nabla_{\theta} \text{Pr}(\tau|\theta) G(\tau) \\ &= \sum_{\tau} \text{Pr}(\tau|\theta) (\nabla_{\theta} \log \text{Pr}(\tau|\theta) G(\tau)) \quad \text{as } \nabla \log f(x) = \frac{\nabla f(x)}{f(x)}\end{aligned}$$

Policy Gradient Framework

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)] \\&= \nabla_{\theta} \sum_{\tau} \text{Pr}(\tau|\theta) G(\tau) \\&= \sum_{\tau} \nabla_{\theta} \text{Pr}(\tau|\theta) G(\tau) \\&= \sum_{\tau} \text{Pr}(\tau|\theta) (\nabla_{\theta} \log \text{Pr}(\tau|\theta) G(\tau)) \quad \text{as } \nabla \log f(x) = \frac{\nabla f(x)}{f(x)} \\&= \mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log \text{Pr}(\tau|\theta) G(\tau)]\end{aligned}$$

Policy Gradient Framework

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [G(\tau)] \\ &= \nabla_{\theta} \sum_{\tau} \text{Pr}(\tau|\theta) G(\tau) \\ &= \sum_{\tau} \nabla_{\theta} \text{Pr}(\tau|\theta) G(\tau) \\ &= \sum_{\tau} \text{Pr}(\tau|\theta) (\nabla_{\theta} \log \text{Pr}(\tau|\theta) G(\tau)) \quad \text{as } \nabla \log f(x) = \frac{\nabla f(x)}{f(x)} \\ &= \mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log \text{Pr}(\tau|\theta) G(\tau)]\end{aligned}$$

Problem: $\text{Pr}(\tau|\theta)$ is unknown

Note that

$$\nabla_{\theta} \log \text{Pr}(\tau|\theta) = \nabla_{\theta} \log \prod_{t=0}^{T-1} \pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t)$$

Note that

$$\begin{aligned}\nabla_{\theta} \log \text{Pr}(\tau|\theta) &= \nabla_{\theta} \log \prod_{t=0}^{T-1} \pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \\ &= \nabla_{\theta} \sum_{t=0}^{T-1} \log \left(\pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \right)\end{aligned}$$

Note that

$$\begin{aligned}\nabla_{\theta} \log \Pr(\tau|\theta) &= \nabla_{\theta} \log \prod_{t=0}^{T-1} \pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \\ &= \nabla_{\theta} \sum_{t=0}^{T-1} \log \left(\pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \right) \\ &= \nabla_{\theta} \sum_{t=0}^{T-1} (\log \pi_{\theta}(a_t|s_t) + \log \mu(s_{t+1}|s_t, a_t))\end{aligned}$$

Note that

$$\begin{aligned}\nabla_{\theta} \log \text{Pr}(\tau|\theta) &= \nabla_{\theta} \log \prod_{t=0}^{T-1} \pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} \log \left(\pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \right) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} (\log \pi_{\theta}(a_t|s_t) + \log \mu(s_{t+1}|s_t, a_t)) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} \log \pi_{\theta}(a_t|s_t) + \cancel{\nabla_{\theta} \sum_{t=0}^{T-1} \log \mu(s_{t+1}|s_t, a_t)}\end{aligned}$$

Note that

$$\begin{aligned}\nabla_{\theta} \log \text{Pr}(\tau|\theta) &= \nabla_{\theta} \log \prod_{t=0}^{T-1} \pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} \log \left(\pi_{\theta}(a_t|s_t) \mu(s_{t+1}|s_t, a_t) \right) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} (\log \pi_{\theta}(a_t|s_t) + \log \mu(s_{t+1}|s_t, a_t)) \\&= \nabla_{\theta} \sum_{t=0}^{T-1} \log \pi_{\theta}(a_t|s_t) + \cancel{\nabla_{\theta} \sum_{t=0}^{T-1} \log \mu(s_{t+1}|s_t, a_t)} \\&= \nabla_{\theta} \sum_{t=0}^{T-1} \log \pi_{\theta}(a_t|s_t)\end{aligned}$$

This gives the core equation for Vanilla Policy Gradient (VPG) or **REINFORCE**

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G(\tau) \right]$$

VPG generates a trajectory, learns, then discards it, generates another.

Clever trick 1: The future cannot affect the past

- In REINFORCE, each $\log \pi_{\theta}(a_t|s_t)$ is multiplied by the full return $G(\tau)$.
- But a_t cannot influence rewards that happened **before** it.
- Can be shown that we can replace $G(\tau)$ with the **reward-to-go**:

$$G_t := \sum_{t'=t+1}^T r_{t'} = r_{t+1} + \dots + r_T$$

- This gives a lower-variance, unbiased estimator.

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G(\tau) \right] \\ &= \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G_t \right]\end{aligned}$$

Algorithm 1 Vanilla Policy Gradient ($\gamma = 1$)

Input: Environment p , Number of episodes M

- 1: **for** episode = 1 to M **do**
 - 2: Generate episode $s_0, a_0, r_1, s_1, \dots, s_{T-1}, a_{T-1}, r_T, s_T$
 - 3: Define $G_t = \sum_{i=t}^{T-1} r_i$ for $0 \leq t \leq T-1$
 - 4: Define gain $J(\boldsymbol{\theta}) = \sum_{t=0}^{T-1} G_t \log \pi_{\boldsymbol{\theta}}(a_t | s_t)$
 - 5: Gradient **ascent** step $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \eta \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
 - 6: **end for**
-

VPG generates a trajectory, learns, then discards it, generates another.

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

$$\mathbb{E}_{x \sim p}[f(x)] = \sum_x p(x) f(x)$$

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

$$\begin{aligned} \mathbb{E}_{x \sim p}[f(x)] &= \sum_x p(x) f(x) \\ &= \sum_x \frac{p(x)}{q(x)} q(x) f(x) \end{aligned}$$

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

$$\begin{aligned} \mathbb{E}_{x \sim p}[f(x)] &= \sum_x p(x) f(x) \\ &= \sum_x \frac{p(x)}{q(x)} q(x) f(x) \\ &= \sum_x q(x) \frac{p(x)}{q(x)} f(x) \end{aligned}$$

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

$$\begin{aligned} \mathbb{E}_{x \sim p}[f(x)] &= \sum_x p(x) f(x) \\ &= \sum_x \frac{p(x)}{q(x)} q(x) f(x) \\ &= \sum_x q(x) \frac{p(x)}{q(x)} f(x) \\ &= \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

Importance Sampling Theorem (IST)

For any two probability distributions p and q where $q(x) > 0$ whenever $p(x) > 0$:

$$\mathbb{E}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right]$$

Proof:

$$\begin{aligned} \mathbb{E}_{x \sim p}[f(x)] &= \sum_x p(x) f(x) \\ &= \sum_x \frac{p(x)}{q(x)} q(x) f(x) \\ &= \sum_x q(x) \frac{p(x)}{q(x)} f(x) \\ &= \mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

The ratio $\frac{p(x)}{q(x)}$ is called the **importance weight** or **importance ratio**.

VPG with Importance Sampling

Problem: VPG is on-policy: we must generate fresh trajectories for each policy update.

Solution: Use importance sampling to reuse old trajectories from $\pi_{\theta_{old}}$.

The **exact** IS-corrected gradient requires the full trajectory probability ratio:

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\frac{\Pr(\tau|\theta)}{\Pr(\tau|\theta_{old})} \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G_t \right] \\ &= \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\prod_{t'=0}^{T-1} \frac{\pi_{\theta}(a_{t'}|s_{t'})}{\pi_{\theta_{old}}(a_{t'}|s_{t'})} \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G_t \right]\end{aligned}$$

Problem: The product of ratios has very *high variance* for long trajectories.

VPG with Importance Sampling (cont.)

Common approximation: Use per-step importance ratios instead:

$$\begin{aligned}\nabla_{\theta} J(\theta) &\approx \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\sum_{t=0}^{T-1} \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G_t \right] \\ &= \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\sum_{t=0}^{T-1} \frac{\nabla_{\theta} \pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} G_t \right] \quad (\text{as } \pi \nabla \log \pi = \nabla \pi)\end{aligned}$$

Why use the ratio at timestep t ? For the term $\nabla_{\theta} \log \pi_{\theta}(a_t|s_t) G_t$:

- Action a_t was sampled from $\pi_{\theta_{old}}(\cdot|s_t)$, but we evaluate $\nabla_{\theta} \log \pi_{\theta}(a_t|s_t)$
- Correct for this mismatch with $\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

What we ignore: State s_t and return G_t also depend on $\pi_{\theta_{old}}$, but when $\pi_{\theta} \approx \pi_{\theta_{old}}$, these effects are small.

VPG with Importance Sampling Algorithm

Input: Environment, policy π_θ , batch size N , learning rate α

Algorithm: Repeat for each iteration:

- ① Collect N episodes in parallel using current policy π_θ :
 - Generate episode $\tau^i = (s_0^i, a_0^i, r_1^i, s_1^i, \dots, s_T^i)$
 - Store log probabilities: $\log \pi_{\text{old}}(a_t^i | s_t^i) = \log \pi_\theta(a_t^i | s_t^i)$
 - Compute returns $G_t^i = \sum_{k=t+1}^T r_k^i$ for all t
 - Store $\{\tau^i, G_t^i, \log \pi_{\text{old}}(a_t^i | s_t^i)\}_{i=1}^N$.
- ② Sample batch $\mathcal{B} = \{\tau^i, G_t^i, \log \pi_{\text{old}}(a_t^i | s_t^i)\}_{i=1}^{|\mathcal{B}|}$.
- ③ Compute importance sampling joy:

$$J(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{t=0}^{T-1} \frac{\pi_\theta(a_t^i | s_t^i)}{\pi_{\text{old}}(a_t^i | s_t^i)} G_t^i$$

- ④ Gradient ascent: $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

Can reuse trajectories, or slice large rollouts into smaller batches.